## MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. Normal to 2 $x - 3y + 1 = 0$ is

   A. $(3; 2)$
   B. $(2; -3)$
   C. $(2; 3)$
   D. $(1; 0)$

2. A Director of $x + 2y = 0$ is

   A. $(-2; 1)$
   B. $(1; 2)$
   C. $(2; 1)$
   D. $(0; 2)$

3. Centre of $x + 2)^2 + (y - 3)^2 = 4$ :

   A. $(2; 3)$
   B. $(-2; -3)$
   C. $(3; -2)$
   D. $(-2; 3)$

4. Rayon of $x^2 + y^2 = 9$ :

   A. $9$
   B. $81$
   C. $3$
   D. $\sqrt{3}$

5. $x^2 + y^2 = -1$ described

   A. a circle
   B. Empty assembly
   C. Origin
   D. a right

## MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
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1. Projected from (3; 4) on the abscissa axis:

   A. $(0; 4)$
   B. $(0; 0)$
   C. $(3; 0)$
   D. $(4; 3)$

2. Distance from (1; 5) to $y = 2$ :

   A. $3$
   B. $5$
   C. $1$

D. $\sqrt{10}$

3. To define $ax + by + c = 0$ as right, it is necessary

A. $a \neq 0$ only

B. $b \neq 0$ only

C. $c \neq 0$

D. $(a; b) \neq (0; 0)$

4. $x^2 + y^2 = 0$ described

A. A circle of radius 1

B. the only point O

C. a right

D. The plan

5. Circle $x^2 + y^2 = 4$ and straight $x = 2$: number of commonalities

A. 0

B. 2

C. 1

D. Infinite

## True or false?

1. The orthogonal project of P on a right is unique.

2. Any equation $x^2 + y^2 + c = 0$ describes a positive radius circle.

3. A right may encounter a circle in three distinct points.

**Personal review**

I give a nonzero normal; I check the projected; I complete the two squares; I distinguish circle, point and empty set.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.